Chow Varieties

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1 Introduction

The study of parameter spaces for algebraic cycles is a foundational cornerstone of algebraic geometry. While the Hilbert scheme successfully parameterizes closed subschemes of a projective variety by keeping track of the scheme-theoretic structure, there is frequently a need to parameterize uniruled or effective algebraic cycles where the nilpotent structure is forgotten and only the formal sum of subvarieties and their multiplicities is retained. The parameter space that achieves this is called **Chow variety**.

This note explores the representability of the Chow functor, tracing its classical roots to its modern functorial formulation. We will outline the construction and representability in characteristic 0 via the Chow form, and subsequently analyze the geometric and algebraic pathologies that arise in characteristic $p > 0$, highlighting why classical approaches fail and how the moduli problem must be adjusted.

1.1 Historical context

The concept of parameterizing varieties of a given dimension and degree dates back to the classical era of algebraic geometry. The foundations were laid in 1937 by W. L. Chow and B. L. van der Waerden in their seminal paper *Zur algebraischen Geometrie IX*, where they introduced the **Chow form** (or often called the Cayley form) to assign a set of homogeneous coordinates to any projective variety.

By associating an algebraic cycle to a specific point in a projective space, they demonstrated that the set of all effective cycles of a fixed dimension and degree forms a closed algebraic set—the Chow variety. In the 1950s and 60s, Grothendieck incorporated the Chow variety into the modern scheme-theoretic framework, notably establishing the **Hilbert-Chow morphism**, which provides a natural map from the Hilbert scheme of closed subschemes to the Chow variety of cycles. However, defining the Chow variety as representing a well-behaved functor of *families of cycles* over arbitrary base schemes remained a delicate problem, leading to extensive modern treatments by mathematicians such as D. Mumford, J. Kollár, and B. Rydh.

1.2 Representability in characteristic 0

In characteristic 0, the representability of the Chow variety can be established robustly using the classical theory of Chow forms and symmetric products. The main steps to construct the Chow variety $\mathrm{Chow}_{d,k}(\mathbb{P}^n)$ (parameterizing cycles of dimension k and degree d in $\mathbb{P}^n$) and prove its representability through following main steps:

- **Step 1: Construction of the** Ch **Map.** We begin by defining a natural transformation from the functor of families of cycles to a projective space. For a given family of cycles, we construct the Chow map (the Ch morphism), which assigns to each cycle its corresponding point in a projective space $\mathbb{P}(V)$. This projective space parameterizes effective Cartier divisors of a specific degree on the relative Grassmannian.

- **Step 2: The Chow Form.** The precise algebraic object underlying the Ch map is the Chow form. For an individual irreducible k-dimensional subvariety $X \subset \mathbb{P}^n$, we consider the incidence correspondence between X and the Grassmannian $\mathbb{G}(n-k-1, n)$. The locus of $(n-k-1)$-planes that intersect X forms a hypersurface in the Grassmannian. The single defining section of this hypersurface is the Chow form F_X. This construction extends additively to formal cycles.

- **Step 3: Construction of the Universal Family.** To show representability, we must construct a universal family of cycles over the closed subscheme defined by the image of the Ch map. This involves constructing a closed subscheme $\mathcal{U} \subset \text{Chow} \times \mathbb{P}^n$ that universally parameterizes the cycles, derived from the universal properties of the incidence variety and the zero-locus of the universal Chow form.

- **Step 4: Completing the Proof.** Finally, we prove that the Ch map factors through a well-defined closed subscheme (the Chow variety) and that pulling back the universal family $\mathcal{U}$ via this map recovers the original family of cycles. In characteristic 0, the complete reducibility of symmetric groups allows the symmetric quotients to behave perfectly, ensuring that this correspondence is both bijective and functorial, thereby completing the proof of representability.

1.3 Obstructions in positive characteristic

When the characteristic of the base field is $p > 0$, the elegant correspondence between algebraic cycles and their Chow forms breaks down, introducing severe obstructions to representability.

- **Purely Inseparable Extensions:** The primary obstruction arises from purely inseparable field extensions. If an algebraic cycle is covered by a family where the generic fiber is geometrically non-reduced (which can happen in characteristic p due to purely inseparable morphisms), the classical Chow form fails to accurately capture the multiplicity of the cycle.

- **Failure of Symmetric Powers:** The functorial correspondence relies heavily on symmetric products to define additions of cycles. In characteristic p, taking the geometric quotient of a scheme by the symmetric group S_d behaves poorly. The map from an affine space to its symmetric product is no longer separable when p divides d, meaning the symmetric tensors no longer accurately represent the "sets of unordered points" in families.

- **Topological vs. Scheme-Theoretic Fibers:** Because the Chow form can evaluate to a p^e-th power for inseparable families, the classical definition only determines the cycle up to a Frobenius twist. The natural Hilbert-Chow morphism Hilb $\to$ Chow may have purely inseparable fibers, meaning the Chow variety, as classically constructed, is functionally closer to a topological space than a space carrying the correct scheme-theoretic moduli structure.

To resolve these obstructions in modern algebraic geometry, the functorial definition of families of cycles must be modified. Instead of relying on symmetric powers, modern proofs (such as those by Rydh) utilize **divided powers** (or operations in the ring of Witt vectors) to keep track of multiplicities correctly, allowing for a well-behaved moduli stack or scheme even in mixed and positive characteristics.

2 Family of Cycles

Recall some basic definitions for algebraic cycles first.

Definition 2.1. *Let X be of finite type over S. A d-dimensional algebraic cycle on X/S is a formal linear combination $\sum_i a_i[V_i]$, where V_i are irreducible closed subschemes of X over a 0-dimension closed subscheme of S. Denote $Z_d(X)$ to be the set of all d-dimensional algebraic cycles on X/S. We say a cycle is effective if $a_i \geq 0$ for all i. In particular, the fundamental cycle $[X] := \sum_i m_i[X_i]$, where X_i are d-dimensional irreducible components of X and $m_i = \mathrm{length}(\mathcal{O}_{X,X_i})$.*

For a finite S-morphism $f : X \to Y$, define pushforward $f_* : Z_d(X) \to Z_d(Y)$ sending $[V]$ to $\deg(V/W)[W]$, where W is the schematic closure of $f(V)$ and $\deg(V/W) = [K(V) : K(W)]$ the extension degree of function fields.

For a flat S-morphism $f : X \to Y$ of relative dimension r, define pullback $f^* : Z_d(Y) \to Z_{d+r}(X)$ sending $[W]$ to $[f^{-1}(W)]$, where $[f^{-1}(W)]$ is the fundamental cycle of $f^{-1}(W)$ with reduced closed subscheme structure. Note that if f is surjective, then f^* is injective.

Let $Z = \sum_i a_i[V_i]$ be a d-dimensional algebraic cycle on X/S, D Cartier divisor on X. Define the degree of Z with respect to D to be $\deg_D Z = \sum_i a_i(V_i \cdot D^d)$, where $V_i \cdot D^d$ is the intersection product.

Assume X is over k. Then for any field extension K/k, $p : X_K := X \times_k K \to X$ is faithfully flat of relative dimension 0. This gives an injective pullback map $p^* : Z_d(X) \to Z_d(X_K)$. For $Z \in Z_d(X_K)$, we say Z defined over a subfield $k \subset F \subset K$ if Z lies in the image of $Z_d(X_F) \to Z_d(X_K)$.

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